

MAJOR PROJECT REGISTRATION FORM

Date:

Project Category	☒ Research oriented	☐ Application oriented	Batch ID:	CSM - A01
	☐ Product oriented	☐ Other (please specify)		
Specify if others:				

Title of the Project:

“Implementation of AI - LLMs in Agriculture: Crop Analytics Using Sentinel-2A Satellite Data”

Abstract

Crop classification, crop health monitoring, and yield estimation are important for agricultural planning, resource management, and food security. Traditional crop monitoring relies heavily on manual field surveys and expert observation, which are time-consuming, costly, and difficult to scale. Satellite-based remote sensing provides an efficient alternative through repeated observations of agricultural regions. Sentinel-2A multispectral satellite data contains 13 optical bands across visible, near-infrared, and shortwave-infrared regions, providing valuable spectral information for identifying crop types and analyzing vegetation conditions.

The proposed project focuses on developing an intelligent crop analytics framework using Sentinel-2A satellite imagery and ground-truth data. The primary stakeholders include agricultural researchers, departments, planners, farmers, and decision-makers who require information about crop distribution, crop condition, and yield-related patterns. Although existing remote-sensing approaches have shown promising results, challenges such as limited ground-truth data, cloud contamination, mixed pixels, similar spectral signatures, and the computational requirements of multi-temporal data can affect performance.

The proposed system will preprocess Sentinel-2A imagery and use ground-truth data for supervised learning. **U-Net will be used as the primary deep-learning model for pixel-level crop segmentation**, while machine-learning classifiers such as Random Forest, Support Vector Machine (SVM), and Naïve Bayes may be explored for comparison. Vegetation indices including NDVI, NDWI, and EVI will be derived to analyze vegetation and water-related conditions. Model performance will be evaluated using accuracy, F1-score, Intersection over Union (IoU), and Dice score, with ground-truth comparison used to assess reliability.

A key contribution is the integration of a **Large Language Model (LLM)-based analytical layer** above the image-processing and classification models. The LLM will not process raw satellite pixels directly. Instead, it will interpret structured outputs such as crop-segmentation results, crop-area information, vegetation-index values, and model results to generate understandable agricultural insights, including crop-pattern, crop-health, and yield-related analysis.

The project scope is limited to satellite-based crop classification, segmentation, vegetation-index analysis, and LLM-assisted interpretation. Exact yield estimation will depend on the availability of suitable ground-truth and yield-reference data. The expected outcome is an integrated and interpretable crop analytics framework combining **Sentinel-2A, U-Net, machine learning, vegetation indices, and LLM-based reasoning** to support data-driven agricultural decision-making.

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References (Category specific):

1. Kramarczyk, P., & Hejmanowska, B. (2023). UNET Neural Network in Agricultural Land Cover Classification Using Sentinel-2. International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences, XLVIII-1/W3, 85–90. DOI: 10.5194/isprs-archives-XLVIII-1-W3-2023-85-2023.
2. Ravirathinam, P., Ghosh, R., Khandelwal, A., Jia, X., Mulla, D., & Kumar, V. (2024). Combining Satellite and Weather Data for Crop Type Mapping: An Inverse Modelling Approach. Proceedings of the 2024 SIAM International Conference on Data Mining, 445–453. DOI: 10.1137/1.9781611978032.52.
3. Wang, L., Bai, Y., Wang, J., Zhou, Z., Qin, F., & Hua, J. (2023). Histogram matching-based semantic segmentation model for crop classification with Sentinel-2 satellite imagery. GIScience & Remote Sensing, 60(1). DOI: 10.1080/15481603.2023.2281142.
4. Qu, T., Wang, H., Li, X., Luo, D., Yang, Y., Liu, J., & Zhang, Y. (2024). A fine crop classification model based on multitemporal Sentinel-2 images. International Journal of Applied Earth Observation and Geoinformation, 134, 104172. DOI: 10.1016/j.jag.2024.104172.
5. Zhou, Y., He, J., Feng, L., Chen, Y., Wu, T., Zhang, X., & Luo, J. (2025). Parcel-Scale Crop Type Classification Using Tile-Slice-Based Spatial-Temporal Convolutional Networks. Journal of Geodesy and Geoinformation Science. DOI: 10.13203/j.whugis20230136.

Project Team Members

S.No	Regd. No.	Name of the Student	Signature
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Project Guide

Name: Dr. T. Kameswara Rao

Signature:

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Date:

INSTRUCTIONS

(No need to submit this page so you don't have to take the print)

Project category

Project categories were mentioned only for the sake of documentation. **All of you must chose 'Research oriented' projects only.** Very few projects would be accepted if the project statements are chosen from Smart India Hackathon (SIH) or from any other reputed platform for that matter. In such cases, you have to write the problem ID mentioned in those platforms as it is in the 'Project category' section. Also provide the link to the problem statement in the 'References' section.

To write Abstract

(Use the following as a mental checklist)

- Background & Problem Statement:** What is the exact real-world problem your project addresses? State it precisely and narrowly not "we are improving healthcare" but "patients in rural areas lack timely access to dermatology consultation."
- Target Users / Stakeholders:** Who exactly faces this problem? A defined persona, not "everyone" e.g. "first-generation college students applying for scholarships."
- Existing Gap & Evidence:** What do people currently do to cope with this problem, and why is it inadequate? Cite any survey, interview, article, or report that confirms the problem is real.
- Related Work / Competitive Landscape:** What are the existing tools/apps/products/papers that partially address this, and how will yours differ or improve on them? (Feeds directly into your References in Section 3.)
- Proposed Approach (overview):** One paragraph describing your approach/technique/system at a high level no implementation detail needed yet.
- Scope & Boundaries:** What will you solve, and explicitly what will you NOT attempt within this project's timeline?
- Expected Outcome / Success Criteria:** How will you know the project succeeded? State one or two measurable, testable outcomes (e.g. "90% of test users complete the task without guidance").

To write category specific references

Research oriented	Cite 1 base paper your problem statement builds on. The paper must be published within the last 3 years (2023 onward as of 2026). Add 2-4 supporting papers from the literature review if relevant.
Application oriented	Identify 2-3 existing, similar applications to use as your benchmark. For each, cite the app/website itself (as a web reference) and, if available, any article or documentation describing it.
Product oriented	Identify 2-3 existing, similar products (hardware/software) to use as your benchmark. Cite the manufacturer's page, datasheet, or a credible review/comparison article for each.